

Yiming Yang

Ph.D. Candidate in ECE department, Michigan Tech

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EDUCATION

Ph.D. in Electrical and Computer Engineering

Aug 2021 - Apr 2026

Michigan Technological University, Houghton, MI, USA

- Dissertation: Robust Autonomous Driving in Winter Weather: Perception and Integrated Control Under Adverse Conditions

Master in Control Science and Engineering

Aug 2017 - Jul 2020

Chang'an University, Xi'an, China

- Thesis: Constrained Optimization and Distributed Model Predictive Control-Based Merging Strategies for Adjacent Connected Autonomous Vehicle Platoons

Bachelor in Automation

Aug 2013 - Jul 2017

Chang'an University, Xi'an, China

- Graduated from the "Excellent Engineer" program

RESEARCH INTERESTS

- Resilient Perception in Adverse Conditions
- Field Robotics & Autonomous Navigation
- Stochastic & Data-Driven Optimal Planning & Control
- Safety-Critical Deployment & Validation

PUBLICATIONS

1. Why Snow Helps: Geometric and Radiometric Enhancements of Vehicles in LiDAR Point Clouds

Under Review.

Yiming Yang, Jeremy P. Bos,

Submitted to IEEE International Workshop on Metrology for Automotive (IEEE MetroAutomotive 2026)

2. Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection

Under Review.

Yiming Yang, Jeremy P. Bos,

Submitted to IEEE/RSJ International Conference on INTELLIGENT ROBOTS & SYSTEMS (IROS)

3. Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion

Under Review

Yiming Yang, Jeremy P. Bos,

Submitted to IEEE/RSJ International Conference on INTELLIGENT ROBOTS & SYSTEMS (IROS)

4. Aggressive Autonomous Control on Snow and Ice

Apr 2025

Yiming Yang, Jeremy P. Bos

SAE International Journal of Advances and Current Practices in Mobility (10.4271/2025-01-8040)

5. Constrained optimization and distributed model predictive control-based merging strategies for adjacent connected autonomous vehicle platoons

Nov 2019

Haigen Min, Yiming Yang, Yukun Fang, Pengpeng Sun, Xiangmo Zhao, 10.1109/ACCESS.2019.2952049

Open-Source Datasets & Tools

1. **ThermalTrack: Multi-modal self-supervised traversability estimation in whiteout conditions**
A dataset of 7,608 synchronized Thermal/VIS frames with ground truth for resilient perception in low-visibility white-out environments. (Associated with [Pub 1])
2. **WADS-3D: LiDAR-based object detection under heavy snow**
Features 2,050 labeled 64-channel LiDAR scans and 30,104 3D bounding boxes, providing a benchmark for autonomous sensing in extreme heavy snowfall. (Associated with [Pub 2])
3. **labelCloud-Enhanced: Advanced Point Cloud Annotation Tool**
Updated utility for high-efficiency 3D labeling featuring AI-assisted pre-labeling, spatial copy-paste logic, and an overhauled viewport with flexible anchor-point rotation.

RESEARCH EXPERIENCE

- Ph.D. Candidate, Michigan Technological University** Aug 2021 – Present
Proposal: Robust 3D Human Pose Estimation in Adverse Winter Conditions Under Review
Lead Author | Sponsor: Sony Research Award Program
 - Proposed a novel Thermal-VIS fusion approach and the first multi-modal dataset for heavy snow to bridge critical gaps in 3D-HPE under low-visibility conditions.
- SAE Autodrive Challenge II** Jan 2023 – Present
Sponsor: General Motors & SAE International
 - Architected the vehicle control interface and ROS-dSPACE communication pipeline via MicroAutoBox III.
 - Developed safety-critical autonomy state machines with robust failure handling and safe-exit mechanisms.
 - Led HIL/VIL validation and authored DVP&R documentation to track system-wide reliability.
- Railroad Crossing Vehicle Warning (RCVW) System** Aug 2021 – Dec 2022
Sponsor: Federal Railroad Administration (U.S. DOT)
 - Led V2X system implementation and created HIL virtual demonstrations for Connected Vehicles.
 - Developed real-world railroad crossing maps and validated the warning application through field testing.
- UAS Data Integration for Transportation Infrastructure** Nov 2021 – Dec 2022
Sponsor: Michigan Dept. of Transportation (MDOT)
 - Curated datasets from UAV imagery (I-96 corridor) to train deep learning counting models.
 - Derived Origin-Destination (OD) metrics and verified system accuracy against state benchmarks.
- Automated Infrastructure Generation (OSM to VISUM/VISSIM)** Aug 2021 – Dec 2022
Research Assistant | Dept. of Civil, Environmental, and Geospatial Engineering
 - Developed automated pipelines to convert OpenStreetMap data into simulation-ready networks for PTV VISUM/VISSIM.
 - Automated network connectivity, zone division, and spline-based connector generation, reducing modeling time from days to minutes.
 - Conducted multi-scale OD analysis using Waymo (micro-level) and Weijo (macro-level) datasets.
- Master's Student, Chang'an University** Aug 2017 – Jul 2020
Auto-driving EV Control & Trajectory Tracking Jul 2018 – Jun 2020
Sponsor: National Key R&D Program / Shaanxi Province Key Projects
 - Designed drive-by-wire algorithms for steering, propulsion, and braking using incremental PID.
 - Deployed trajectory tracking (Pure Pursuit/MPC) and engineered safety-critical remote-control override systems.
 - Managed the gasoline-to-EV architecture transition and developed an MCU-based vehicle controller.
- QLTD Intelligent Connected Expressway Testbase** Apr 2019 – Sep 2020
Sponsor: QiLu Transportation Development Group
 - Implemented V2X-based adaptive IDM and Forward Collision Warning systems using DSRC/C-V2X.
 - Validated V2V functionalities through real-world highway platoon tests and remote-control system deployment.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Jan 2023 – Present

Course: *Robotic Systems Enterprise (Michigan Tech)*

- Mentored 20+ undergraduates across the **Controls** and **Safety & Testing** teams; managed sprints using SMART goals and provided technical support.
- **Controls Focus:** Guided deployment of steering/propulsion/brake/lighting control and trajectory-tracking algorithms using Model-Based Design (MBD) and Stateflow autonomy state machines.
- **Safety Focus:** Directed functional safety validation (SOTIF) and lateral/longitudinal controller testing for autonomous functions.

TALKS & PRESENTATIONS

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|---|----------|
| 1. Aggressive Autonomous Control on Snow and Ice | Apr 2025 |
| Conference proceedings talk, World Congress Experience (WCX), 2025, Detroit, MI, USA | |
| 2. Deep Deterministic Policy Gradient based Cooperative Platoon Logitudinal Control Strategy | Jan 2021 |
| Conference presentation, Transportation Research Board (TRB), 2021, Virtual | |
| 3. Research on Automatic Emergency Line Change Control Strategy Based on MPC | Jun 2019 |
| Conference presentation, World Transport Convention (WTC), 2019, Beijing, China | |

PATENTS

1. **Autonomous vehicle controllers** (NO: ZL 2019 2 0786357.5)
2. **Autonomous vehicle platoon controller** (NO: ZL 2019 2 0784716.3)
3. **Unmanned vehicle power system and unmanned vehicle** (NO: ZL 2019 2 1037928)

SELECTED HONORS & AWARDS

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| 1. 2nd Place, Concept design report & SRS - SAE/GM AutoDrive Challenge II (Year 4) | Jun 2025 |
| 2. 2nd Place, Highway Challenge - SAE/GM AutoDrive Challenge II (Year 2) | Jun 2023 |
| 3. 3rd Place, Overall Dynamics - SAE/GM AutoDrive Challenge II (Year 2) | Jun 2023 |
| 4. Excellent graduate of Chang'an University | Jun 2020 |
| 5. Excellent graduate of Chang'an University | Jun 2020 |
| 6. First-class Academic Scholarship 2017-2018 | Nov 2018 |
| 7. 2017-2018 University-level Outstanding Graduate Student | Nov 2018 |
| 8. Best Technology Award of 2018 i-VISTA Auto Driving Challenge | Aug 2018 |
| 9. First Prize of 2018 i-VISTA Auto Driving Challenge - Urban Traffic Scene Challenge | Aug 2018 |
| 10. Second Prize of 2018 China Robot Competition - General Service Robot Project | Aug 2018 |

SKILLS

Computational: Python, C/C++, MATLAB/Simulink for algorithm development and data analysis
Control Systems: Real-time control design and implementation (dSPACE, ConfigurationDesk, ControlDesk)
Traffic Simulation: PTV Vissim, PTV Visum
Robotics: ROS, GAZEBO, dSPACE integration with ROS
Hardware: CAN bus analysis, CAD (SOLIDWORKS), and PCB design (Altium Designer)